19: Odds, Odds Ratios, and Logistic Regression

Self-paced tutorial

! Today's plan

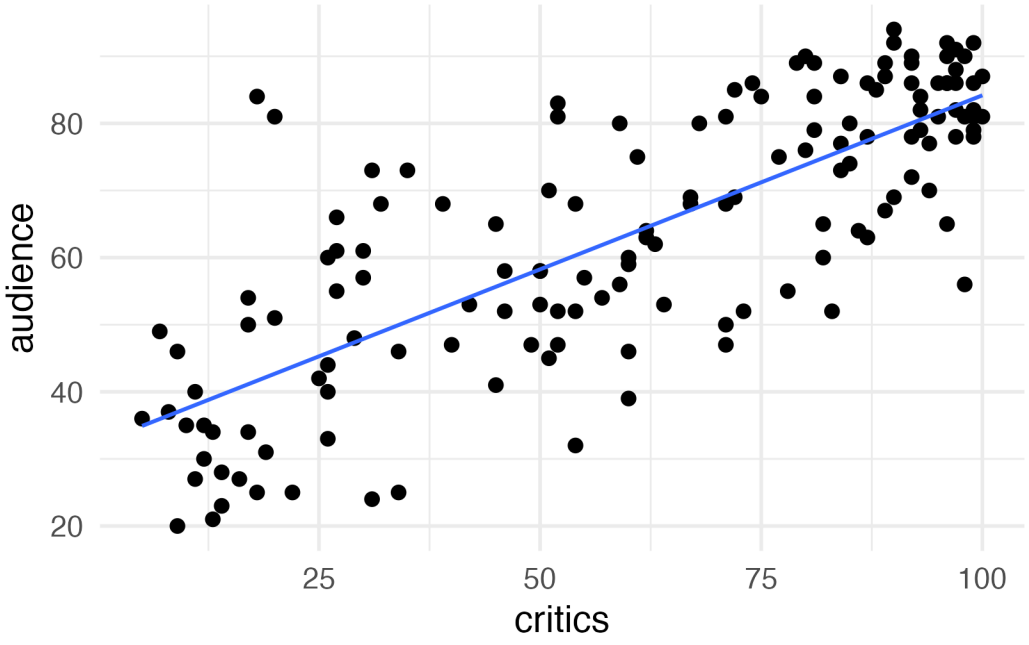
I've lost my voice! 🤐 Instead of a lecture, you'll work through this tutorial with your group at your own pace. It covers the same material as the Day 19 slides and the Day 19 activity.

- **Code chunks:** you can use these as usual
- **Check-in questions:** click an answer to see whether you're right (and why). Try to answer *before* you click!
- **Numbered questions (Q1, Q2, ...):** remember write your answers in your notes.

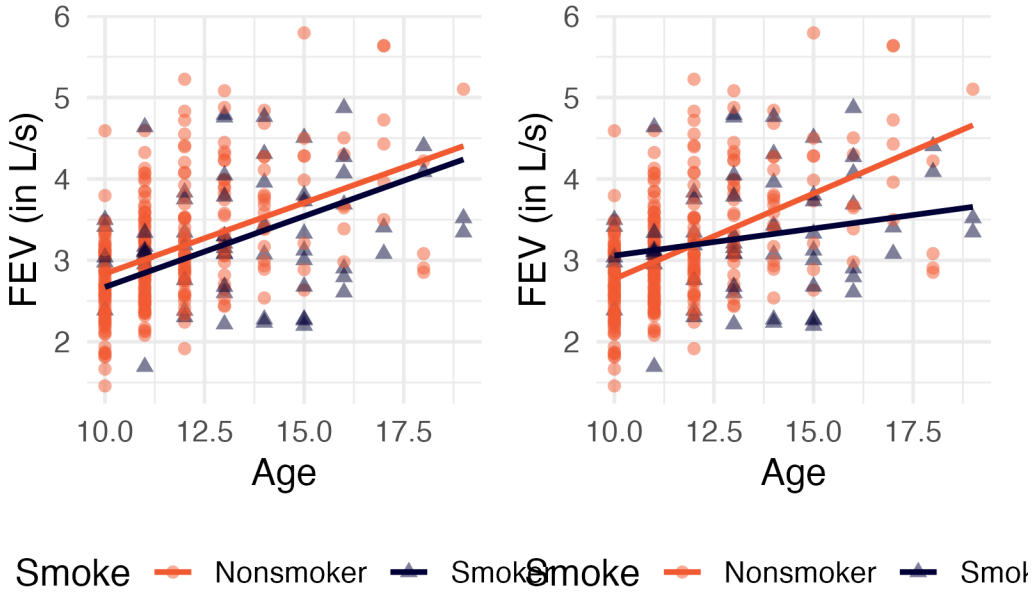
I will wander around and be here to answer any questions you have!

1 Part 0: What we've done so far

1.1 1. SLR: Quantitative outcome and one quantitative predictor

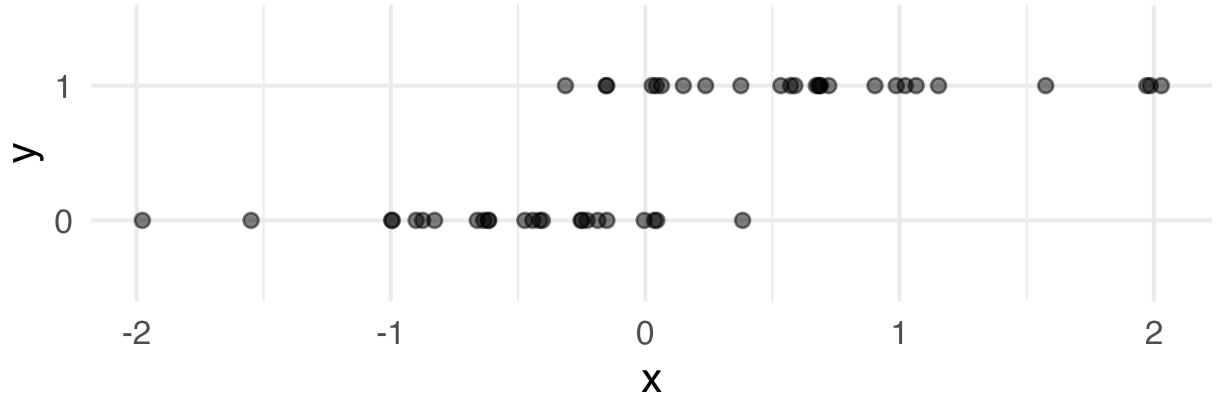


1.2 2. MLR: Quantitative outcome with quantitative and categorical predictors:



1.3 3. Today: a binary outcome

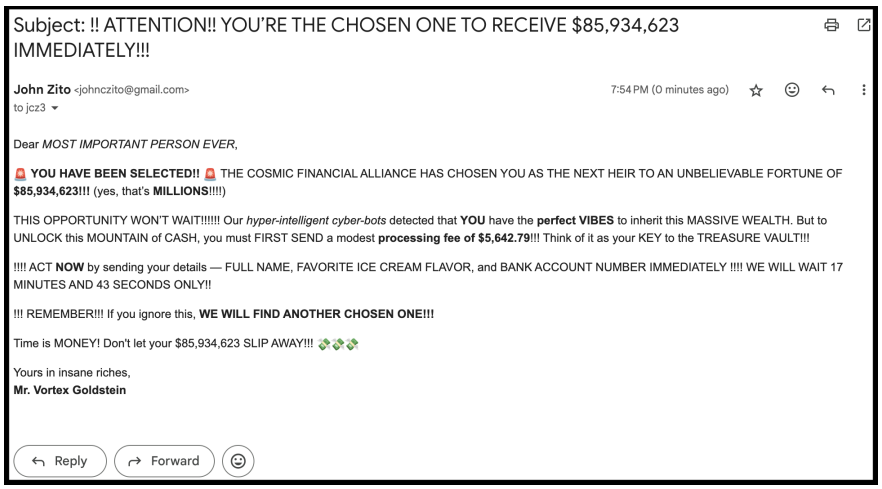
$$y = \begin{cases} 1 & \text{eg. Yes, Win, True, Heads, Success} \\ 0 & \text{eg. No, Lose, False, Tails, Failure} \end{cases}$$



If we can model the relationship between predictors (x) and a binary outcome (y), we can use the model to do a special kind of prediction called *classification*.

1.3.1 Example: Is the e-mail spam or not?

x : word and character counts in an e-mail.


Subject: !! ATTENTION!! YOU'RE THE CHOSEN ONE TO RECEIVE $85,934,623 IMMEDIATELY!!!

John Zito <johnczito@gmail.com> 7:54 PM (0 minutes ago)
to jcz3

Dear *MOST IMPORTANT PERSON EVER*,

🚨 **YOU HAVE BEEN SELECTED!!** 🚨 THE COSMIC FINANCIAL ALLIANCE HAS CHOSEN YOU AS THE NEXT HEIR TO AN UNBELIEVABLE FORTUNE OF **$85,934,623!!!** (yes, that's **MILLIONS**!!!!)

THIS OPPORTUNITY WON'T WAIT!!!!!! Our *hyper-intelligent cyber-bots* detected that **YOU** have the **perfect VIBES** to inherit this MASSIVE WEALTH. But to UNLOCK this MOUNTAIN of CASH, you must FIRST SEND a modest **processing fee of $5,642.79**!!! Think of it as your KEY to the TREASURE VAULT!!!

!!!! ACT **NOW** by sending your details — FULL NAME, FAVORITE ICE CREAM FLAVOR, and BANK ACCOUNT NUMBER IMMEDIATELY !!!! WE WILL WAIT 17 MINUTES AND 43 SECONDS ONLY!!

!!! REMEMBER!!! If you ignore this, **WE WILL FIND ANOTHER CHOSEN ONE!!!**

Time is MONEY! Don't let your $85,934,623 SLIP AWAY!!! 💸💸💸

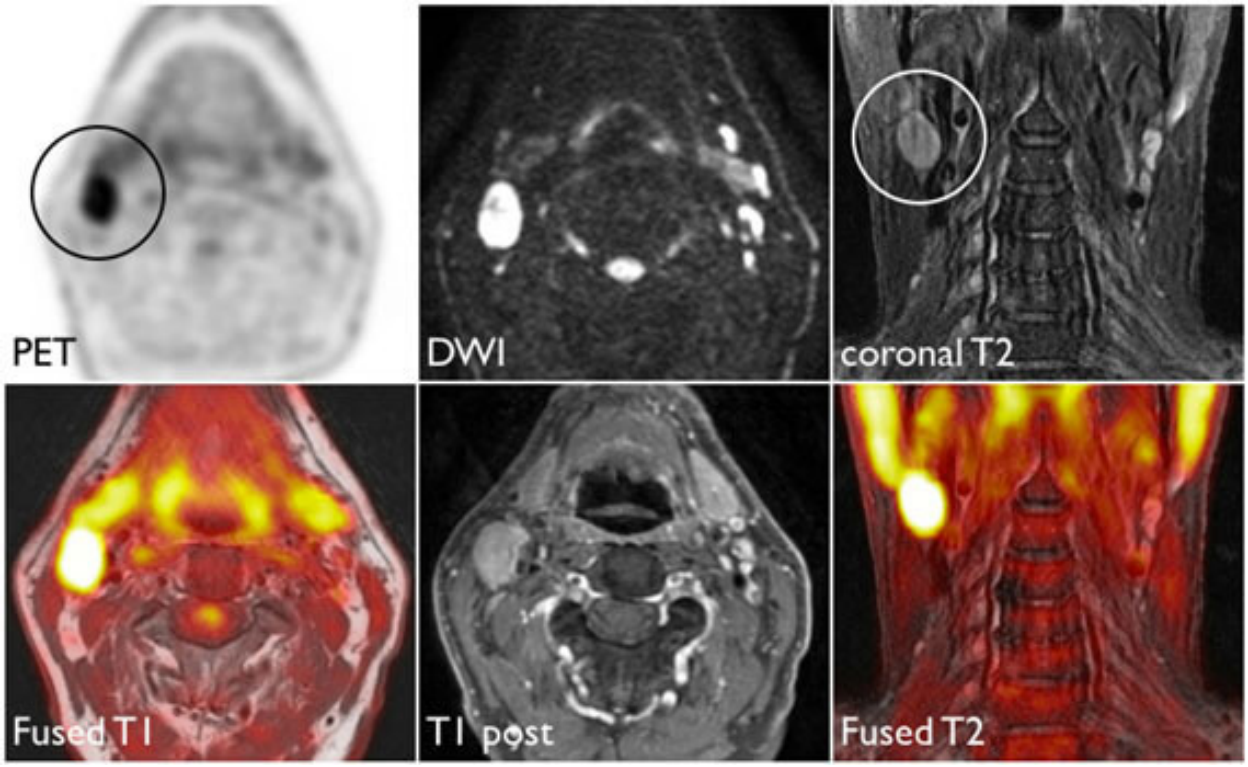
Yours in insane riches,
Mr. Vortex Goldstein

Reply Forward

$$y = \begin{cases} 1 & \text{it's spam} \\ 0 & \text{it's legit} \end{cases}$$

1.3.2 Example: Is it cancer or not?

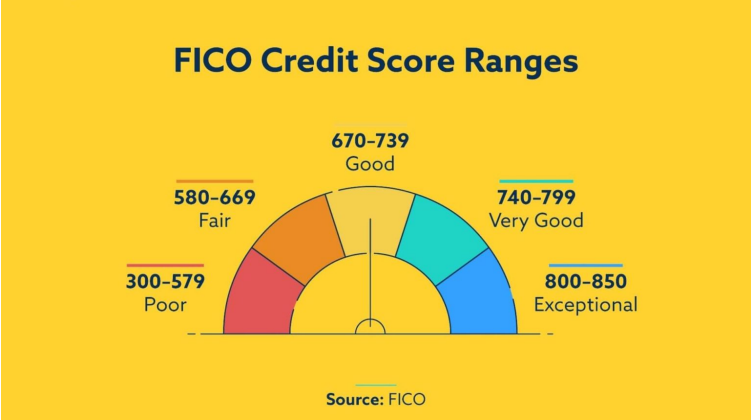
$\mathbf{x}$: features in a medical image.



$$y = \begin{cases} 1 & \text{it's cancer} \\ 0 & \text{it's healthy} \end{cases}$$

1.3.3 Example: Will they default?

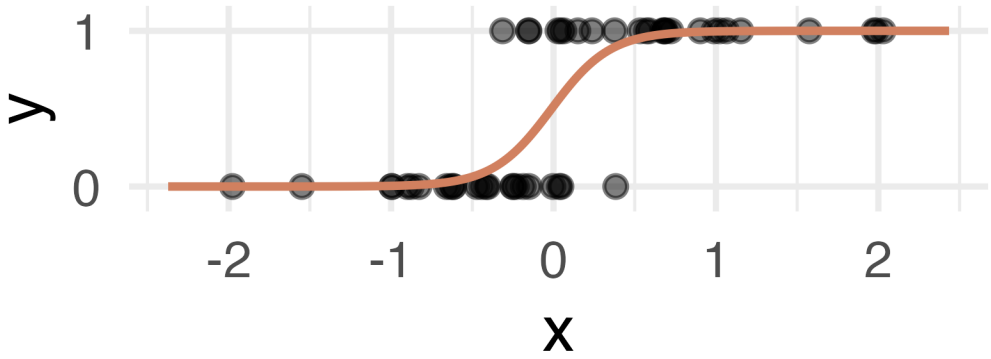
$\mathbf{x}$: financial and demographic info about a loan applicant.



$$y = \begin{cases} 1 & \text{applicant is at risk of defaulting on loan} \\ 0 & \text{applicant is safe} \end{cases}$$

1.3.4 Modeling probabilities instead of responses

This data is hard to model! Instead of modeling y directly, we model the **probability** that $y = 1$:



So, our prediction questions turn into:

- "Given new email, what's the probability that it's spam?' '
- "Given new image, what's the probability that it's cancer?' '
- "Given new loan application, what's the probability that they default?' '

Why don't we model y directly?

- **Recall regression with a numerical outcome**:

  - Our models do not output *guarantees* for y, they output predictions that describe behavior *on average*

- **Similar when modeling a binary outcome**:

  - Our models cannot directly guarantee that y will be zero or one. The correct analog to "on average" for a 0/1 outcome is "what's the probability?"

Today, we'll work through an example with the goal of predicting whether an article is fake news or not.

2 Part 1: The data

This data set has information on 150 news articles, some that are real and some that are fake. Our goal is to build a statistical model to explain `type` based on features of the article.

- `type`: "real" or "fake" / `type_num`: 0 (real) or 1 (fake)
- `title` and `url` of article
- `title_words`: how many words in the title
- `title_has_excl`: whether title has an exclamation point
- `title_caps_percent`: percent of title in capital letters
- `negative`: percent of words that have negative sentiment
- `positive`: percent of words that have positive sentiment

Run the code chunk below to load the data. (Every chunk after this one needs `news`, so start here!)

```
#| autorun: true
library(ggformula)
news <- read.csv("https://aluby.github.io/stat230/data/fake_news_bayesrules.csv")
head(news[, c("type", "type_num", "title_has_excl", "title_caps_percent", "title_word
```

Q1. Before we do anything else, how do you think `title_has_excl` and `title_caps_percent` are related to `type_num`? (Write your answer down in your notes)

3 Part 2: Probability, odds, and odds ratios

3.1 Probability vs. odds

If π is the probability of an event, then the **odds** of the event are

$$\text{odds} = \frac{\pi}{1-\pi} = \frac{P(\text{event})}{P(\text{not event})}$$

Probability π	Odds $\pi/(1-\pi)$	How we'd say it
0.5	1	"even odds"
0.75	3	"3 to 1"
0.2	0.25	"1 to 4"

Probabilities live between 0 and 1. Odds live between 0 and ∞.

3.2 Fake news and exclamation points

Here's a table of article type by whether the title has an exclamation point:

```
table(news$title_has_excl, news$type) |>
  addmargins()
```

For titles **with** an exclamation point (the TRUE row):

- $P(\text{fake} \mid \text{exclamation}) = 16/18 \approx 0.889$
- $\text{Odds}(\text{fake} \mid \text{exclamation}) = \dfrac{16/18}{2/18} = \dfrac{16}{2} = 8$

Notice the shortcut: the odds are just (number fake) / (number real). You can use the chunk below as a calculator.

```
16 / 18   # probability
16 / 2    # odds
```

3.3 Odds ratios

To compare the two groups, we divide their odds. This is the **odds ratio (OR)**:

$$OR = \frac{\text{Odds}(\text{fake} \mid \text{exclamation})}{\text{Odds}(\text{fake} \mid \text{no exclamation})} = \frac{8}{0.5} = 16$$

Q2. What would an odds ratio of exactly 1 mean in this context? What about an odds ratio less than 1?

4 Part 3: Why not linear regression?

Now let's use a quantitative predictor: `title_caps_percent`. Here's what happens if we fit our usual linear regression model to a 0/1 response. We'll treat $\hat{y}$ as the probability of a 1.

```
gf_jitter(type_num ~ title_caps_percent, data = news, width = 0.2, height = 0.05) |>
  gf_lm(color = "#f15a31") |>
  gf_labs(y = "Fake = 1")

lm_fit <- lm(type_num ~ title_caps_percent, data = news)
predict(lm_fit, newdata = data.frame(title_caps_percent = c(0, 43.75)))
```

5 Part 4: The logistic regression model

The trick is to model a version of the probability that isn't stuck between 0 and 1: the **log-odds** (also called the **logit**).

$$\log\left(\frac{\pi(x)}{1-\pi(x)}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$$

where $\pi(x) = P(Y = 1 \mid x)$.

Scale	Formula	Range
Probability	π	0 to 1
Odds	$\frac{\pi}{1-\pi}$	0 to ∞
Log-odds	$\log\left(\frac{\pi}{1-\pi}\right)$	$-\infty$ to ∞

Because the log-odds can be any number, it's OK for the right-hand side to be a line.

5.1 Fitting the model in R

We use `glm()` with `family = binomial`. The syntax is otherwise the same as `lm()`.

```
news_glm <- glm(type_num ~ title_caps_percent, data = news, family = binomial)
coef(news_glm)
```

So the fitted model is

$$\log\left(\frac{\hat{\pi}(x)}{1-\hat{\pi}(x)}\right) = -0.932 + 0.098x$$

where x is `title_caps_percent`.

6 Part 5: Interpreting $\hat{\beta}_1$

$\hat{\beta}_1 = 0.098$ is the expected change in the **log-odds** of being fake for a 1-unit increase in `title_caps_percent`. But we don't think in log-odds! Exponentiating both sides gives the odds:

$$\widehat{\text{odds}}(x) = e^{\hat{\beta}_0 + \hat{\beta}_1 x} = e^{\hat{\beta}_0}\, e^{\hat{\beta}_1 x}$$

If x increases by 1, the odds get *multiplied* by

$$\frac{\widehat{\text{odds}}(x+1)}{\widehat{\text{odds}}(x)} = \frac{e^{\hat{\beta}_0} e^{\hat{\beta}_1 (x+1)}}{e^{\hat{\beta}_0} e^{\hat{\beta}_1 x}} = e^{\hat{\beta}_1}$$

So $e^{\hat{\beta}_1}$ is an **odds ratio**.

```
exp(coef(news_glm))
```

7 Part 6: From log-odds to probabilities

What's the probability that an article with **20%** capital letters in the title is fake? Work from the inside out:

1. **Log-odds:** $\hat{\eta} = -0.932 + 0.098(20) \approx 1.023$
2. **Odds:** $\hat{\omega} = e^{\hat{\eta}} = e^{1.023} \approx 2.78$
3. **Probability:** $\hat{\pi} = \dfrac{\hat{\omega}}{1+\hat{\omega}} = \dfrac{e^{\hat{\eta}}}{1+e^{\hat{\eta}}} \approx 0.736$

```
b <- unname(coef(news_glm))
eta <- b[1] + b[2] * 20       # log-odds
odds <- exp(eta)              # odds
prob <- odds / (1 + odds)     # probability
c(log_odds = eta, odds = odds, prob = prob)
```

> **Tip**
>
> To go from odds to probability: $\pi = \dfrac{\text{odds}}{1+\text{odds}}$. For example, odds of 3 ("3 to 1") is a probability of $3/4 = 0.75$.

7.1 Letting R do it: `predict()`

`predict()` works for `glm()` models too, but be careful about the scale:

```
predict(news_glm, newdata = data.frame(title_caps_percent = 20))
predict(news_glm, newdata = data.frame(title_caps_percent = 20), type = "response")
```

7.2 The fitted model on all three scales

Run the chunk below to see the same fitted model on all three scales.

```
x <- seq(0, 60, by = 0.5)
b <- coef(news_glm)
log_odds <- b[1] + b[2] * x
three_scales <- data.frame(
  x = rep(x, 3),
  value = c(log_odds, exp(log_odds), exp(log_odds) / (1 + exp(log_odds))),
```

```
  scale = rep(c("1. log-odds", "2. odds", "3. probability"), each = length(x))
)
gf_line(value ~ x, data = three_scales, color = "#003300", linewidth = 1) |>
  gf_facet_wrap(~scale, scales = "free_y") |>
  gf_labs(x = "title_caps_percent", y = "")
```

Q3. Describe the shape of the fitted model on each of the three scales. Why does the probability curve flatten out near 1?

8 Part 7: Connecting back to odds ratios

What if the predictor is the 0/1 variable `title_has_excl`?

```
excl_glm <- glm(type_num ~ title_has_excl, data = news, family = binomial)
coef(excl_glm)
exp(coef(excl_glm))
```

Q4. What does $e^{\widehat{\beta}_0} = 0.5$ represent in context?

9 Part 8: Your turn

Now, fit a logistic regression model using all three variables we've discussed today (`title_has_excl`, `title_caps_percent`, and `title_words`) and report the `summary()`.

```
news_glm_2 <- glm(type_num ~ ___ + ___ + ___, data = news, family = binomial)
summary(___)
```

Q5. Write out the fitted model (on the log-odds scale) for articles (A) with an exclamation point in the title and (B) without an exclamation point in the title.

Q6. Write out interpretations for all slope coefficients in the model in terms of the **odds** of being fake news.

Q7. Use `predict()` to find the estimated probability that an article with an exclamation point, 10% capital letters, and 12 words in the title is fake.

```
# your code here
```

Q8. Based on what we know about OLS (**o**rdinary **l**east **s**quares, which includes both SLR and MLR), which predictors appear to be most important for this model?

Q9. What questions do you have about the `summary()` output? (Write them down! We'll talk about them next class.)

9.1 Function quick reference

Function	Purpose
`table(x, y) |> addmargins()`	Two-way table with row and column totals
`glm(formula, data, family = binomial)`	Fit a binary logistic regression model
`coef(model), exp(coef(model))`	Coefficients on the log-odds scale; odds ratios
`summary(model)`	View model summary and Wald tests
`predict(model, newdata)`	Predictions on the log-odds scale
`predict(model, newdata, type = "response")`	Predictions on the probability scale